

Causal DAGs: the structural meaning of CI

PS 200C, Spring 2026

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Where we are, and today's question

So far: **selection on observables**. We assumed CI:

$$\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i \mid X_i$$

Then if we condition on the right X , treatment is as-if random within strata.

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DAGs visualize and tell us how to remove unwanted sources of association.

Confounding, with numbers

Suppose people self-select into a new drug. You find:

	<i>Recovered</i> (count / total)	
	Drug	No drug
Mild	81 of 87 (93%)	234 of 270 (87%)
Severe	192 of 263 (73%)	55 of 80 (69%)
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With POMs, we might have noted that $\mathbb{E}[Y(0)|D = 1] - \mathbb{E}[Y(0)|D = 0] < 0$ (negative bias)

Preview: Confounding, with a DAG

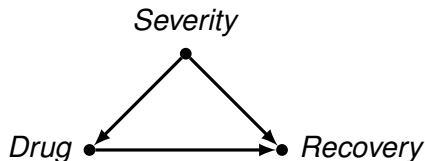
What we *believe* about the world:

- Severity increases drug-taking
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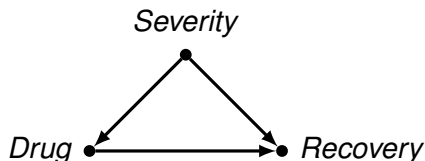
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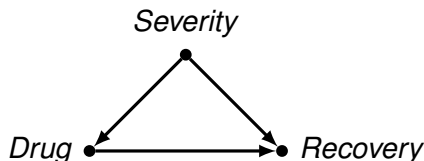
Now you can see the two reasons *Drug* and *Recovery* are associated:

- 1 $Drug \rightarrow Recovery$ (what we want),
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Conditioning on *Severity* removes (2), leaving only (1).

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Three things a DAG is *not*:

- Not a list of the variables in your dataset. (Many nodes may be unobserved.)
- Not a statistical model. (No parametric assumptions.)
- Not a complete theory of the world. (Just enough structure for the causal claim you want to make.)

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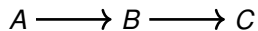
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Our task: figure out, for each, (a) when is A associated with C ? (b) what happens when we condition on B ?

Chain

$A \rightarrow B \rightarrow C$:



Chain

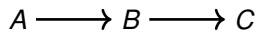
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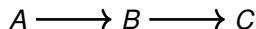
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$A \rightarrow B \rightarrow C$:



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Moral: a chain transmits association; conditioning on the middle blocks it.

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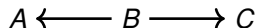
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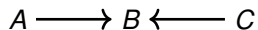
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Moral: a common cause creates spurious association; conditioning on it removes it. **This is classical confounding.**

Collider (common effect)

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Moral: colliders are the reverse of the other two. They are *already blocked*, and conditioning on one *creates* an association that wasn't there.

Colliders are real

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Lesson: Conditioning on something can give rise to a spurious association...this will matter when we start trying to isolate causal effects.



d-separation

Putting the three patterns together:

Rule

***d*-separation.** *A and C are d-separated by a set Z iff every path between them is blocked. A path is blocked when:*

- *It contains a chain or fork at a node in Z, **or***
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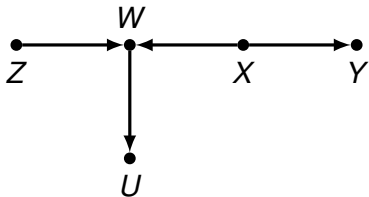
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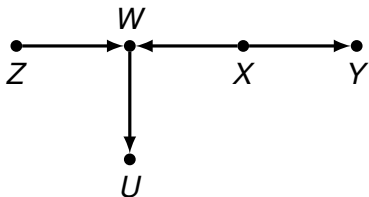
In plain English: When A and C are d-separated by Z in this way, you can condition on Z as a way to remove the association between A and C due to those paths.

Practice: d -separation



Looking at Z and Y :

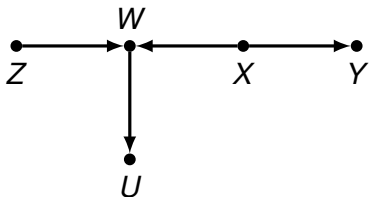
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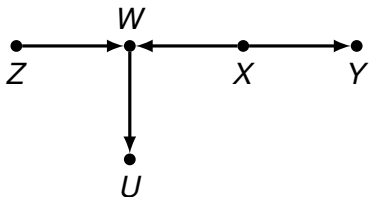
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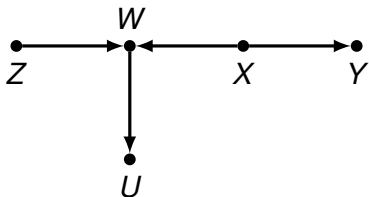
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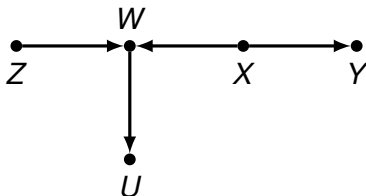
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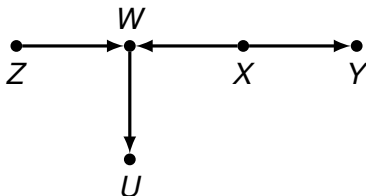
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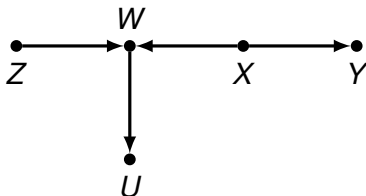
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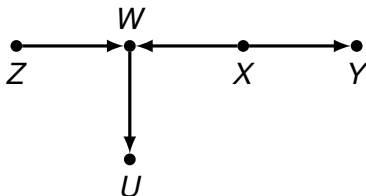
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- d -separated by $\{U\}$? **No** — U is a descendant of collider W , so conditioning on it also opens the path.

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That rule is the **backdoor criterion**.

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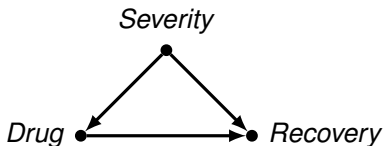
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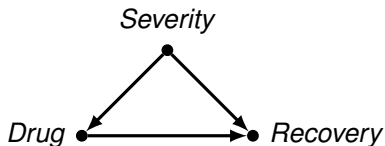
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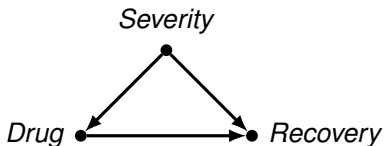
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Frontdoor: $Drug \rightarrow Recovery$.

Backdoor: $Drug \leftarrow Severity \rightarrow Recovery$. Condition on *Severity* to block it.

The backdoor criterion

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- 2 leaves *causal* paths untouched (no descendants of D),

The backdoor criterion

Rule

Backdoor criterion. $\{X\}$ satisfies the backdoor criterion relative to (D, Y) if:

- 1 No node in X is a descendant of D (don't block frontdoor paths), **and**
- 2 X blocks every backdoor path from D to Y .

For humans, find a set X that:

- 1 blocks all *spurious* paths (every backdoor),
- 2 leaves *causal* paths untouched (no descendants of D),
- 3 doesn't *open new* spurious paths (no colliders or their descendants unless paired with another blocker downstream).

Backdoor = CI: two languages, one object

Theorem (informal)¹: if X satisfies the backdoor criterion in a DAG, then

$$\{Y_{1i}, Y_{0i}\} \perp\!\!\!\perp D_i \mid X_i$$

in any SCM consistent with that DAG.

¹For formal statements: Pearl, Glymour & Jewell (2016) *Primer* Thm 4.3.1; Shpitser, VanderWeele & Robins (2010) adjustment criterion.

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We can then apply the “adjustment formula” if we wanted to recover $P(Y)$ as if we “made everyone have treatment $D = d$ ”, i.e. $P(Y \mid do(d))$:

$$P(Y \mid do(d)) = \sum_x P(Y \mid D=d, X=x) P(X=x).$$

Or, since we have been working with expectations:

$$\mathbb{E}(Y \mid do(d)) = \sum_x \mathbb{E}(Y \mid D=d, X=x) P(X=x)$$

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Adjustment formula for ATE

If you use this to get the ATE, $\mathbb{E}[Y \mid do(D = 1)] - \mathbb{E}[Y \mid do(D = 0)]$, you see exactly what we say with adjustment under SOO:

$$\sum_x (\mathbb{E}[Y \mid D=1, X=x] - \mathbb{E}[Y \mid D=0, X=x]) P(X=x)$$

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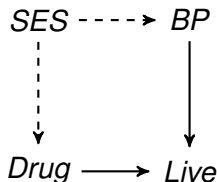
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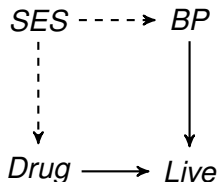
As before, sub-stratification is the literal way to do so, but in practice we'll often need to use other approaches that give us an estimate as if $p(X|D = 1)$ and $p(X|D = 0)$.

Example 1: a hidden confounder, but a workable adjustment set



SES unobserved (dashed). Can we identify the effect of *Drug* on *Live*?

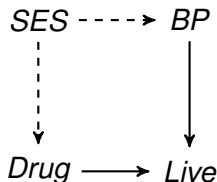
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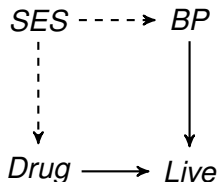
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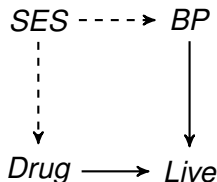
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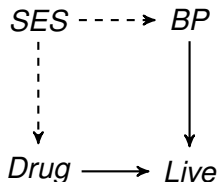
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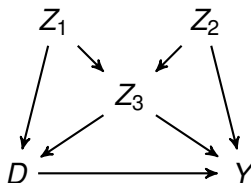
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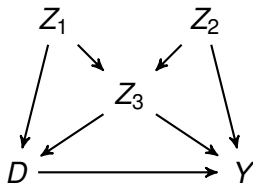
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Example 2: multiple valid adjustment sets

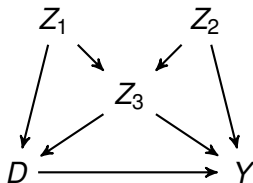


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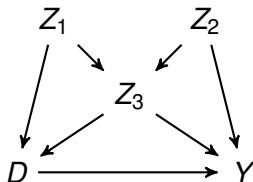
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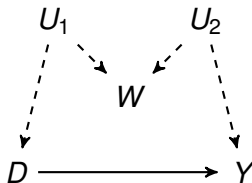
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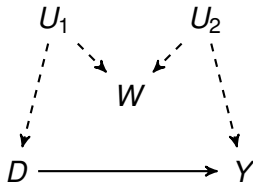


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- *Invariance*: all give you the same estimand in the non-parametric sense. Different estimators in practice, though.

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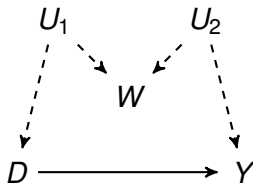


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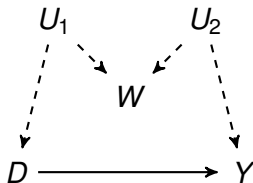
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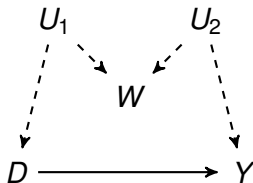
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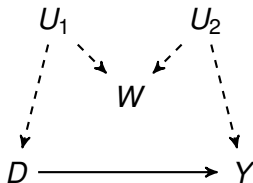
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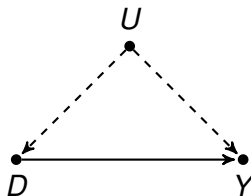
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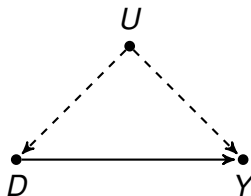
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Lessons: 1. DAGs help you see if CI holds; and 2. Conditioning on extra variables can *break* an identification that was fine without them.

Example 4: Failure of identification

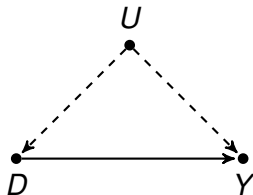


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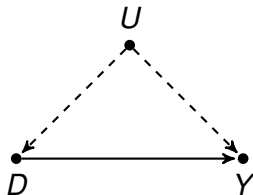
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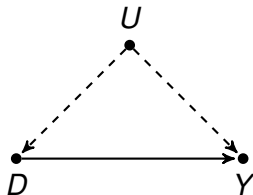
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What to do? SOO+sensitivity, or find another ID strategy.

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This is a feature not a bug: the DAG is surfacing key disagreement that needs to be resolved or what-if'd out.

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Online courses / lectures:

- *Causal Diagrams: Draw Your Assumptions Before Your Conclusions* — Miguel Hernán, EdX.
- *A Crash Course in Causality* — Jason Roy, Coursera.
- *Richard McElreath, “Science Before Statistics: Causal Inference.”
<https://www.youtube.com/watch?v=KNPYUVmY3NM>
- *Richard McElreath, *Statistical Rethinking* lecture series on YouTube (more on DAGs + Bayesian modeling).

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- **The graph tells you when you're stuck.** Example 4 is the honest payoff: identification isn't guaranteed.